

The circular mixing threshold is rational: $\mathfrak{M}_c(G)$ has reduced numerator at most $|V(G)| + 1$

github.com/graph-theory-AI

Machine-generated note

18 September 2026

Abstract

For every finite loopless graph G with n vertices and at least one edge, we prove that its circular mixing threshold belongs to the finite set of rationals at least 2 whose reduced numerator is at most $n + 1$. Edgeless graphs have threshold 1 under the source definition. Hence the circular mixing threshold is always rational; the proof uses integer difference constraints to show that mixing can change only at finitely many rational breakpoints.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 6](#) records the full provenance.

1 Introduction

For integers $k, q > 0$ with $k \geq 2q$, a (k, q) -colouring of a graph G is a map $f : V(G) \rightarrow \{0, \dots, k - 1\}$ such that

$$q \leq |f(u) - f(v)| \leq k - q \quad (uv \in E(G)).$$

Let $\mathcal{R}_{k,q}(G)$ be the graph whose vertices are these colourings, with two adjacent when they differ at exactly one vertex. The graph G is (k, q) -mixing when this colour graph is connected. Brewster and Noel define [2, Definition 1.2]

$$\mathfrak{M}_c(G) = \inf \{r \in \mathbb{Q} : r \geq \chi_c(G) \text{ and } G \text{ is } (k, q)\text{-mixing whenever } k/q \geq r\}.$$

The OpenProblemGarden entry [1] asks verbatim:

Question 1.1. Is $\mathfrak{M}_c(G)$ always rational?

The source paper gives bounds relating $\mathfrak{M}_c$ to the ordinary mixing threshold [2, Theorem 4.12], but leaves rationality open. Later work gives a complexity dichotomy for fixed circular targets [3] and characterizes mixing when the ratio is below 4 [4]; neither settles [Question 1.1](#).

For $N \geq 2$, define the finite set

$$\mathcal{E}_N = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}_{>0}, 2b \leq a \leq N \right\}.$$

Membership is unchanged by reducing the fraction, and every member has reduced numerator at most N .

Repair. The catalog rendering omits the source definition's condition $r \geq \chi_c(G)$, while the original writeup instead declared an empty colour graph to be non-mixing. For a graph with an edge the two formulations are equivalent: if $r \geq \chi_c(G)$, every ratio at least r admits a colouring. If $2 \leq r < \chi_c(G)$, a rational circular ratio between them admits none; if $r < 2 = \chi_c(G)$, the pair $(2, 1)$ itself is non-mixing for every graph with an edge. Here we use Brewster–Noel's definition throughout. For an edgeless graph $\chi_c(G) = 1$ and every colour graph is connected, so $\mathfrak{M}_c(G) = 1$; the original writeup's suggestion of $-\infty$ came only from the truncated rendering.

2 Statement and proof architecture

Theorem 2.1 (Finite set of circular mixing thresholds). *Let G be a finite loopless graph on n vertices. If G has an edge, then*

$$\mathfrak{M}_c(G) \in \mathcal{E}_{n+1}.$$

If G is edgeless, then $\mathfrak{M}_c(G) = 1$. In particular, $\mathfrak{M}_c(G)$ is always rational; in the nonempty case its reduced numerator is at most $n + 1$.

Idea of proof. Orient every edge from the smaller to the larger colour. A fixed orientation class is the integer-point set of a difference-constraint system with weights $-q$ and $k - q$. Away from ratios in $\mathcal{E}_n$, every such class is connected. Its feasibility, and the feasibility of a transition between two classes, can change only when k/q crosses a member of $\mathcal{E}_{n+1}$. Hence mixing is constant on every component interval outside that finite set. All ratios at least $n + 1$ mix, so the supremum of the bad ratios is a member of $\mathcal{E}_{n+1}$ and equals $\mathfrak{M}_c(G)$. [\[full proof\]](#)

3 Difference constraints and orientation classes

Lemma 3.1 (Unit-step connectivity). *Let*

$$P = \{x \in \mathbb{Z}^s : 0 \leq x_i \leq K, x_j - x_i \leq c_{ij} \text{ for } (i, j) \in A\},$$

where all data are integral. If $P \neq \emptyset$ and the weighted directed graph A has no directed simple cycle of total weight zero, then the graph on P obtained by changing one coordinate by 1 at a time is connected.

Idea of proof. The set is closed under coordinatewise minimum. To descend from x to $\min(x, y)$, decrease a coordinate above the target. If every such coordinate were blocked by a tight outgoing constraint, following blocking arcs would give a tight cycle of total weight zero. [\[full proof\]](#)

Every colouring f induces an orientation D_f by directing $u \rightarrow v$ when $f(u) < f(v)$. For an orientation D , let $X_D(k, q)$ be the colourings inducing D . For every arc $u \rightarrow v$ these are exactly the integral solutions of

$$x_u - x_v \leq -q, \quad x_v - x_u \leq k - q, \quad 0 \leq x_v \leq k - 1.$$

Lemma 3.2 (Exceptional ratios and feasibility). *Let D be an orientation on s vertices.*

- (i) *If $k/q \notin \mathcal{E}_s$, every nonempty $X_D(k, q)$ is connected by unit single-vertex recolourings.*
- (ii) *Whether $X_D(k, q)$ is nonempty depends only on k/q , and is constant on each component interval of $(2, \infty) \setminus \mathcal{E}_s$.*

Idea of proof. A simple cycle in the variable constraint graph has weight $ak - bq$ with $b \leq s$, so zero weight forces $k/q = b/a \in \mathcal{E}_s$; part (i) follows from [Lemma 3.1](#). For feasibility, add a reference vertex encoding the box $0 \leq x_i \leq k - 1$. The resulting system is feasible exactly when it has no negative cycle. Cycles through the reference vertex have weight $ak - bq - 1$, and integrality turns nonnegativity into the strict ratio comparison $k/q > b/a$. [\[full proof\]](#)

4 The finite quotient and eventual mixing

For fixed k, q , let $\mathcal{Q}_{k,q}(G)$ have as vertices the orientations D with $X_D(k, q) \neq \emptyset$. Join D and D' when some colourings in their classes differ at one vertex.

Proposition 4.1 (Local constancy of mixing). *Let G have n vertices and let I be a component interval of $(2, \infty) \setminus \mathcal{E}_{n+1}$. Then either G is (k, q) -mixing for every positive integer pair with $k/q \in I$, or it is mixing for none of them.*

Idea of proof. Class feasibility uses [Lemma 3.2](#) on n variables. A transition changing vertex v is tested by splitting v into two nonadjacent twins, one with the old and one with the new incident orientations; this is a feasibility test on $n + 1$ variables. Thus $\mathcal{Q}_{k,q}(G)$ is constant throughout I . As I also avoids $\mathcal{E}_n$, every class is connected, so the colour graph is connected exactly when the quotient is. [\[full proof\]](#)

Lemma 4.2 (Eventual mixing). *Every n -vertex graph is (k, q) -mixing whenever $k/q \geq n + 1$.*

Idea of proof. The $n + 1$ colours $0, q, \dots, nq$ are pairwise compatible. Within its orientation class, any colouring reaches an injective colouring from this palette; a spare palette colour then connects all injective colourings. [\[full proof\]](#)

5 Remarks

Remark 5.1 (Scope). The theorem proves rationality, but it does not prove that $\mathfrak{M}_c(G)$ is integral or that the threshold is attained. The related circular mixing number $\mathfrak{m}_c(G)$ is not treated here. The theorem concerns finite loopless graphs; parallel edges would only duplicate constraints.

Remark 5.2 (Computational audit). The referee reimplemented the quotient algorithm by Bellman–Ford feasibility tests and compared it with brute-force recolouring on 360 parameter pairs across 12 graphs, with zero mismatches. No representation-dependent ratio or violation of local constancy was found. The observed thresholds include

$$\mathfrak{M}_c(K_3) = 4, \quad \mathfrak{M}_c(K_4) = 5, \quad \mathfrak{M}_c(K_5) = 6, \quad \mathfrak{M}_c(C_5) = \mathfrak{M}_c(C_6) = \mathfrak{M}_c(C_7) = 4.$$

The values for K_3, K_4, K_5 show that the index $n + 1$ in the finite exceptional set cannot in general be replaced by n .

6 Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The proof was found by `gpt-6-astra`, reasoning effort `max`, in a single pass started on 2026-09-16, for OpenProblemGarden catalog id `mixing_circular_colourings_0`, campaign `leg_attacks_opg`. (2) An independent adversarial referee agent (`claude-opus-5`) re-derived the argument, checked the source definition and later literature, and ran the computations described above on 2026-09-17; its verdict was `MINOR GAPS`. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-17 `claude-fable-5-1` began the rewrite but left only a compilable skeleton containing the preamble, title and section headings. (4) On 2026-09-18 `Codex` completed the present note and moved the full proofs to [Appendix A](#). It repaired the sole substantive gap by proving equivalence with Brewster–Noel’s actual definition, restored the omitted $r \geq \chi_c(G)$ clause, corrected the edgeless case from the original writeup’s truncated-definition discussion, stated the elementary inclusion $\mathcal{E}_n \subseteq \mathcal{E}_{n+1}$ where used, and added the source and later-work citations. No other mathematical content was changed. **No human mathematician has checked the argument.**

A Full proofs

Lemma 3.1 (Unit-step connectivity). *Let*

$$P = \{x \in \mathbb{Z}^s : 0 \leq x_i \leq K, \ x_j - x_i \leq c_{ij} \text{ for } (i, j) \in A\},$$

where all data are integral. If $P \neq \emptyset$ and the weighted directed graph A has no directed simple cycle of total weight zero, then the graph on P obtained by changing one coordinate by 1 at a time is connected.

Proof. The set P is closed under coordinatewise minimum. Indeed, for $x, y \in P$ put $z_i = \min(x_i, y_i)$. If $z_i = x_i$, then $z_j - z_i \leq x_j - x_i \leq c_{ij}$; if $z_i = y_i$, use y instead. The box constraints are also preserved.

It therefore suffices to connect x to $z = \min(x, y)$ by coordinate decreases. At a current point, still denoted $x \geq z$, put $D = \{i : x_i > z_i\}$. Decreasing x_i by one cannot violate the lower box bound. The only possible obstruction is a tight outgoing constraint $x_j - x_i = c_{ij}$. Its endpoint j must lie in D : otherwise $x_j = z_j$, and since $x_i \geq z_i + 1$ we would have

$$z_j - z_i \geq x_j - (x_i - 1) = c_{ij} + 1,$$

contrary to $z \in P$.

If every $i \in D$ were blocked, choose one tight blocking arc from each. Repeatedly following these arcs inside the finite set D produces a directed simple cycle. Summing the tight equalities around it telescopes to show that its total weight is zero, contrary to the hypothesis. Hence some coordinate in D may be decreased. Integral slack ensures that every nontight inequality remains valid after the unit change. Repeating strictly decreases $\sum_i (x_i - z_i)$ until z is reached. Apply the same argument to y and concatenate the two paths. $\square$

Lemma 3.2 (Exceptional ratios and feasibility). *Let D be an orientation on s vertices.*

- (i) *If $k/q \notin \mathcal{E}_s$, every nonempty $X_D(k, q)$ is connected by unit single-vertex recolourings.*
- (ii) *Whether $X_D(k, q)$ is nonempty depends only on k/q , and is constant on each component interval of $(2, \infty) \setminus \mathcal{E}_s$.*

Proof. For each arc $u \rightarrow v$ of D , the two inequalities defining $X_D(k, q)$ give arcs of weights $-q$ and $k - q$ in the variable constraint graph. A directed simple cycle of length b containing a arcs of weight $k - q$ has total weight

$$a(k - q) + (b - a)(-q) = ak - bq.$$

If it has weight zero, then $a \geq 1$ and $k/q = b/a$. Since $b \leq s$ and $k/q \geq 2$, this ratio lies in $\mathcal{E}_s$. Thus when $k/q \notin \mathcal{E}_s$, Lemma 3.1 makes every nonempty $X_D(k, q)$ connected, proving (i).

For (ii), add a reference vertex $*$ and arcs $* \rightarrow i$ of weight $k - 1$ and $i \rightarrow *$ of weight 0. With $x_* = 0$, these impose exactly $0 \leq x_i \leq k - 1$. An integral system of difference constraints is feasible if and only if its weighted digraph has no negative directed cycle. Necessity follows by summing constraints. For sufficiency, shortest-path distances from $*$ are finite in the absence of a negative cycle, are integral, satisfy every constraint, and the incident arcs place them in the required box.

It is enough to inspect simple cycles. A cycle avoiding $*$ has weight $ak - bq$ with $1 \leq b \leq s$ and $0 \leq a \leq b$. When $a = 0$ it is always negative; otherwise its nonnegativity is the weak comparison $k/q \geq b/a$. A cycle through $*$ uses its two incident arcs exactly once and has weight

$$ak - bq - 1, \quad 0 \leq b \leq s - 1, \quad 1 \leq a \leq b + 1.$$

Because $ak - bq$ is an integer,

$$ak - bq - 1 \geq 0 \iff ak - bq > 0 \iff k/q > b/a.$$

Thus feasibility is a finite Boolean combination of weak and strict comparisons of k/q with fixed rationals. Every comparison point in $[2, \infty)$ belongs to $\mathcal{E}_s$. This proves both ratio dependence and constancy on the complementary intervals. $\square$

Proposition 4.1 (Local constancy of mixing). *Let G have n vertices and let I be a component interval of $(2, \infty) \setminus \mathcal{E}_{n+1}$. Then either G is (k, q) -mixing for every positive integer pair with $k/q \in I$, or it is mixing for none of them.*

Proof. By Lemma 3.2, the vertex set of $\mathcal{Q}_{k,q}(G)$ is determined by feasibility tests on n variables. We now express each possible edge as a similar test on $n + 1$ variables. Suppose D, D' differ only on edges incident with a vertex v ; otherwise no one-vertex recolouring can connect them. Replace v by two nonadjacent twins v_0, v_1 , each with neighbourhood $N_G(v)$. Orient the edges at v_0 as in D , the edges at v_1 as in D' , and all remaining edges in their common direction.

A colouring respecting this orientation gives the common colours away from v , the old colour at v_0 , and the new colour at v_1 , hence a transition. Conversely each transition gives such a colouring. Since $D \neq D'$, some incident edge reverses; its inequalities force the two twin colours to differ. Thus the edge test is exactly orientation-class feasibility on this $(n + 1)$ -vertex graph.

Throughout I , all vertex and edge tests are constant by Lemma 3.2; hence $\mathcal{Q}_{k,q}(G)$ itself is constant. Since $\mathcal{E}_n \subseteq \mathcal{E}_{n+1}$, the interval I avoids $\mathcal{E}_n$, and every nonempty orientation class is connected. A connected quotient path lifts by moving within a class to the endpoint of each next transition. Conversely, projecting a recolouring path gives a walk in the quotient. Therefore $\mathcal{R}_{k,q}(G)$ is nonempty and connected exactly when $\mathcal{Q}_{k,q}(G)$ is, proving the claim. $\square$

Lemma 4.2 (Eventual mixing). *Every n -vertex graph is (k, q) -mixing whenever $k/q \geq n + 1$.*

Proof. Assume $k \geq (n + 1)q$. The palette $C = \{0, q, 2q, \dots, nq\}$ lies in $\{0, \dots, k - 1\}$ and any two distinct palette colours differ by at least q and at most $nq \leq k - q$. Thus every injective map $V(G) \rightarrow C$ is a (k, q) -colouring.

For an arbitrary colouring f , the induced orientation D_f is acyclic because colours strictly increase along each arc. A topological ordering of D_f , assigned distinct palette colours in increasing order, gives an injective palette colouring in $X_{D_f}(k, q)$. Since every element of $\mathcal{E}_n$ is at most n , the ratio $k/q \geq n + 1$ lies outside $\mathcal{E}_n$; Lemma 3.2(i) therefore connects f to that palette colouring.

All injective maps into C are connected. To reach a target, fix vertices one at a time. If the desired colour of the next vertex is occupied by an unfixed vertex, move that blocker first to an unused colour. There is always an unused colour because $|C| = n + 1 > |V(G)|$. Every intermediate map remains injective and hence proper. Combining these paths proves mixing. $\square$

Theorem 2.1 (Finite set of circular mixing thresholds). *Let G be a finite loopless graph on n vertices. If G has an edge, then*

$$\mathfrak{M}_c(G) \in \mathcal{E}_{n+1}.$$

If G is edgeless, then $\mathfrak{M}_c(G) = 1$. In particular, $\mathfrak{M}_c(G)$ is always rational; in the nonempty case its reduced numerator is at most $n + 1$.

Proof. First suppose that G has an edge. Define the bad-ratio set

$$B_G = \{r \in \mathbb{Q}_{\geq 2} : \text{for some representation } r = k/q, G \text{ is not } (k, q)\text{-mixing}\}.$$

The existential quantifier allows arbitrary representation dependence at exceptional ratios. We have $2 \in B_G$: if G is nonbipartite it has no $(2, 1)$ -colouring, while if it is bipartite, flipping the two colours on a nontrivial component cannot be done by single-vertex recolourings. By Lemma 4.2, $B_G \cap [n + 1, \infty) = \emptyset$.

By Proposition 4.1, on each component interval of $(2, \infty) \setminus \mathcal{E}_{n+1}$, either every rational belongs to B_G or none does. At the finitely many points of $\mathcal{E}_{n+1}$ arbitrary behaviour is harmless. Therefore B_G is a union of selected rational open intervals between consecutive exceptional points, together with a subset of the exceptional points. It is nonempty and bounded, and its supremum is either an included exceptional point or the right endpoint of a selected interval. Hence

$$\sigma := \sup B_G \in \mathcal{E}_{n+1}.$$

For every rational $\ell > \sigma$, every pair with $k/q \geq \ell$ mixes; for every $\ell < \sigma$, some bad ratio exceeds ℓ . Thus the infimum in the universal mixing condition is σ , regardless of behaviour at equality. Moreover

$\sigma \geq \chi_c(G)$. Indeed, if $\chi_c(G) > 2$, admissible rational ratios approach $\chi_c(G)$ from below and are bad because they admit no colouring; if $\chi_c(G) = 2$, then $2 \in B_G$. As explained in the repair in the introduction, for thresholds at least $\chi_c(G)$ the nonempty formulation is exactly Brewster–Noel’s. Therefore $\mathfrak{M}_c(G) = \sigma \in \mathcal{E}_{n+1}$.

If G is edgeless, $\chi_c(G) = 1$ by the standard convention and changing one vertex at a time connects every two colourings. The source definition then gives $\mathfrak{M}_c(G) = 1$. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the *original* writeup. Its step numbers refer to that writeup, not to this note. The scripts it mentions are in `verification_astra/scripts/mixing_circular_colourings_0/`. Treat it as machine output, not as an authority.

B.1 Interpretation

Source definitions (verified against the actual paper, arXiv:1412.3493, §1, Definition 1.2):

$$\text{mc}(G) := \inf\{k/q \in \mathbb{Q} : k/q \geq \chi_c(G) \text{ and } G \text{ is } (k, q)\text{-mixing}\} \quad \text{Mc}(G) := \inf\{r \in \mathbb{Q} : r \geq \chi_c(G) \text{ and } G \text{ is } (k, q)\text{-mixing for all } k/q \geq r\}$$

and, earlier in §1,

“Given graphs G and H such that $G \rightarrow H$, define the H -colour graph of G , denoted $C_H(G)$, to be the graph on vertex set $\text{HOM}(G, H)$ where $f \sim g$ if $f(v) \neq g(v)$ for exactly one $v \in V(G)$. The graph G is said to be H -mixing if $C_H(G)$ is connected.” ...
 “Thus, G is k -mixing or (k, q) -mixing if the corresponding colour graph is connected.”

Two conventions therefore matter, and the OpenProblemGarden rendering in `prompt.md` **drops** the clause $r \geq \chi_c(G)$ from the definition of Mc . The writeup follows OPG, and compensates by declaring (§0) that “mixing means that this graph is nonempty and connected”, and by taking the infimum over all rationals ℓ .

I checked that the two readings define the same number, which the writeup does **not** do:

- Let $S_{\text{src}} = \{r \in \mathbb{Q} : r \geq \chi_c(G) \text{ and } \forall (k, q) \text{ with } k/q \geq r, C_{\{k, q\}}(G) \text{ is connected}\}$; let $S_{\text{wr}} = \{r \in \mathbb{Q} : \forall (k, q) \text{ with } k/q \geq r, C_{\{k, q\}}(G) \text{ is non-empty and connected}\}$.
- For $r \geq \chi_c(G)$: $k/q \geq r \geq \chi_c(G)$ implies $G \rightarrow G_{\{k, q\}}$ (standard: $G \rightarrow G_{\{k, q\}} \Leftrightarrow k/q \geq \chi_c(G)$), and χ_c is rational and attained for finite G , so non-emptiness is automatic and the two conditions coincide.
- For $r < \chi_c(G)$: $\chi_c(G)$ is rational, so there is a rational $k/q \in [r, \chi_c(G))$; then G has no (k, q) -colouring, so $r \notin S_{\text{wr}}$; and $r \notin S_{\text{src}}$ by the explicit clause. So $S_{\text{src}} = S_{\text{wr}}$ and the infima agree.

Hence the writeup’s object is Brewster–Noel’s $\text{Mc}(G)$, and OPG’s Question (“Is $\text{Mc}(G)$ always rational?”) is the one being answered. The interpretation is fair, not a strawman: the writeup proves a *strictly stronger* statement (reduced numerator $\leq n+1$) than mere rationality, and does not lean on any literal-reading loophole. Its one interpretation-driven contortion — the paragraph claiming edgeless graphs would give $\text{Mc} = -\infty$ “if the displayed definition is read literally” — is an artefact of OPG’s truncation only; under the real Definition 1.2 an edgeless graph has $\chi_c = 1$ and $\text{Mc} = 1$, which is rational with numerator $1 \leq n+1$, so the excluded case is not actually an exception. This is over-caution, not gaming.

Scope note: Brewster–Noel’s Question 1 in §7.1 reads “Is the circular mixing threshold (**number**) always rational?”, i.e. it covers both Mc and mc . The writeup only treats Mc . (Its machinery does give $\text{mc} \in E_{\{n+1\}}$ by the identical argument — the infimum of a finite union of open rational intervals with endpoints in $E_{\{n+1\}}$ plus a subset of $E_{\{n+1\}}$ again lies in $E_{\{n+1\}}$ — but this is not stated or proved.)

Notation used below: $E_N := \{a/b : a, b \in \mathbb{Z}_{\{>0\}}, 2b \leq a \leq N\}$ = the (finite) set of rationals ≥ 2 whose *reduced* numerator is at most N .

B.2 Step-by-step findings

#	Step	Label	Note
1	Statement of E_N ; Theorem $\text{Mc}(G) \in \text{VALID}_{E_{\{n+1\}}}$; “reduced numerator $\leq n+1$ ”		If $a/b \geq 2$ with $a \leq N$, the reduced numerator is $\leq a \leq N$. E_N is finite, $\max E_N = N$.
2	Lemma 1 , closure of $\mathbb{P}$ under coordinatewise min	VALID	With $z = \min(x, y)$: if $z_i = x_i$ then $z_j \setminus \setminus z_i \setminus \leq x_j \setminus \setminus x_i \setminus \leq c_{ij}$; symmetric otherwise. Box constraints preserved.

#	Step	Label	Note
3	Lemma 1 , “blocked $\implies$ blocking $j \in \mathbb{D}$ ”	VALID	If $x_j = z_j$ and $x_i \geq z_i + 1$, then $z_j \setminus z_i \geq x_j \setminus (x_i \setminus 1) = c_{ij} + 1$, contradicting $z \in \mathbb{P}$. Correctly re-derived.
4	Lemma 1 , “all of $\mathbb{D}$ blocked $\implies$ zero-weight cycle”	VALID	Iterating the blocking map $\beta : \mathbb{D} \rightarrow \mathbb{D}$ yields a <i>simple</i> directed cycle of tight arcs; summing $x_j - x_i = c_{ij}$ around it telescopes to $0 = \Sigma c$. Contradiction. Descent terminates since $\Sigma(x_i - z_i)$ strictly decreases. Also correct that only <i>out</i> -arcs of i can block a decrease of x_i , and that the lower box bound cannot, since $x_i > z_i \geq 0$.
5	§2: $X_{\mathbb{D}}(k, q) =$ integer points of the difference system $x_u - x_v \leq -q$, $x_v - x_u \leq k - q$ (for $u \rightarrow v$ in $\mathbb{D}$), $0 \leq x \leq k-1$	VALID	Matches the standard $G_{\{k, q\}}$ definition ($q \leq f(u) - f(v) \leq k - q$, linear not modular). $x_u - x_v \leq -q$ forces $x_u < x_v$, so every feasible point induces <i>exactly</i> $\mathbb{D}$; conversely every colouring inducing $\mathbb{D}$ is feasible. Equality of the two sets, hence the partition, is correct.
6	§2: a simple directed cycle has weight $ak - bq$; zero weight $\implies k/q = b/a \in \mathbb{E}_n$	VALID	a arcs of weight $k - q$, $b - a$ of weight $-q$, total $a(k - q) - (b - a)q = ak - bq$. $a = 0$ gives $-bq < 0 \neq 0$, so $a \geq 1$. $b \leq n$ (simple cycle in an n -vertex digraph). $k/q \geq 2$ forces $b \geq 2a$, hence $b/a \in \mathbb{E}_n$.
7	Lemma 2 ($k/q \notin \mathbb{E}_n \implies$ every non-empty $X_{\mathbb{D}}$ is connected by unit recolourings)	VALID	Immediate from 4+6. Unit moves are legal single-vertex recolourings and all intermediates are genuine (k, q) -colourings. Verified computationally (see below).
8	Lemma 3 : $*$ -augmentation encodes $0 \leq x_i \leq k-1$; feasibility $\iff$ no negative cycle; integrality of shortest-path distances	VALID	$x_* = 0$, arcs $* \rightarrow i$ of weight $k-1$ and $i \rightarrow *$ of weight 0 give exactly the box. All vertices reachable from $*$; with integer weights and no negative cycle the distances are finite integers satisfying every constraint, and lie in $[0, k-1]$. Restricting to simple cycles is legitimate (any negative closed walk contains a negative simple cycle).
9	Lemma 3 : taxonomy of simple cycles; $*$ -free cycles $ak - bq$, $1 \leq b \leq s$, $0 \leq a \leq b$	VALID	Re-derived. The $a = 0$ case ($\mathbb{D}$ contains a directed cycle) is <i>always</i> negative, i.e. a ratio-independent obstruction — so it does not break the claimed constancy.
10	Lemma 3 : $*$ -cycles have weight $ak - bq - 1$ with $0 \leq b \leq s-1$, $1 \leq a \leq b+1$	VALID	A simple $*$ -cycle is $* \rightarrow i \rightarrow j \rightarrow *$; the internal path has a' arcs of weight $k - q$ and b' of weight $-q$ with $a' + b' \leq s-1$. Setting $a = a' + 1$, $b = a' + b'$ gives exactly the stated ranges.
11	Lemma 3 : the integrality trick $ak - bq - 1 \geq 0 \iff ak - bq > 0 \iff k/q > b/a$	VALID	$ak - bq \in \mathbb{Z}$, so $\geq 1 \iff > 0$; $a \geq 1$ so division is legitimate. This is the load-bearing step that removes all dependence on the <i>scale</i> of (k, q) (the $k-1$ box bound is the only place k entered non-homogeneously). Verified computationally: the set of feasible orientation classes never depended on the representation of k/q .

#	Step	Label	Note
12	Lemma 3 conclusions (1) ratio-only, (2) constant on components of $(2, \infty) \setminus E_{-s}$	VALID	All comparison points are b/a with $b \leq s$; if $b/a \geq 2$ then $b/a \in E_{-s}$, and if $b/a < 2$ the comparison is constantly true on $(2, \infty)$. On a component interval no comparison point is interior, so weak and strict comparisons agree and are constant.
13	§4: definition of $Q_{\{k, q\}}(G)$	VALID	Vertex set = non-empty orientation classes; edges = existence of a one-vertex-apart pair. Well defined.
14	§4: twin-graph test for $D \sim D'$	VALID	If two colourings differ only at v , then D, D' agree off v . The twin graph (v_0, v_{-1}) non-adjacent, each with $N(v)$; edges at v_0 oriented by D , at v_{-1} by D' , the rest by their common direction) is simple and loopless on $n+1$ vertices, and its feasible points biject with the desired transitions. $x_{\{v_0\}} \neq x_{\{v_{-1}\}}$ is automatic: $D \neq D'$ means some incident edge is reversed, forcing $x_{\{v_0\}} < x_u < x_{\{v_{-1}\}}$ (or the reverse). So the $n+1$ in the theorem is exactly where it comes from.
15	§4: $R_{\{k, q\}}(G)$ non-empty+connected $\iff Q_{\{k, q\}}(G)$ non-empty+connected, when $k/q \notin E_{-n}$	VALID	$(\Leftarrow)$ lift a Q -path using Lemma 2 inside each class; $(\Rightarrow)$ project an R -path (consecutive colourings have equal or Q -adjacent orientations). Non-emptiness transfers trivially.
16	Proposition 4 (mixing constant on each component I of $(2, \infty) \setminus E_{\{n+1\}}$)	VALID	Verified computationally end-to-end. Vertex tests are Lemma-3 tests on n vertices, edge tests on $n+1; E_{-n} \subseteq E_{\{n+1\}}$, so I avoids both exceptional sets and $Q_{\{k, q\}}(G)$ is literally the same finite graph throughout I . The $E_{-n} \subseteq E_{\{n+1\}}$ inclusion is used but not stated; trivial.
17	§5: palette $C = \{0, q, \dots, nq\}$ is a clique of $G_{\{k, q\}}$ when $k/q \geq n+1$	VALID	$k \geq (n+1)q$ gives $nq \leq k-q \leq k-1$ (palette inside range) and any two palette colours differ by between q and $nq \leq k-q$.
18	§5: topological order of the acyclic D_{-f} gives an injective palette colouring in $X_{\{D_{-f}\}}$; Lemma 2 connects f to it	VALID	D_{-f} acyclic since colours strictly increase along arcs; assigning increasing palette colours along a topological order reproduces exactly D_{-f} . $k/q \geq n+1 \implies k/q \notin E_{-n}$ since every element of E_{-n} is $\leq n$.
19	§5: injective palette maps are mutually reachable; hence $k/q \geq n+1 \implies (k, q)$ -mixing	VALID	$n+1$ colours vs n vertices leaves a free colour at every stage; the “fix one vertex at a time, evict the blocker to a free colour” argument is correct and all intermediates are injective, hence proper. No computational counterexample found.
20	§6: $2 \in B_G$ for a graph with an edge	VALID	$(2, 1)$ -colourings are proper 2-colourings. Non-bipartite: none exist (empty $\implies$ not mixing under the stated convention). Bipartite with an edge: every vertex in a non-trivial component is frozen, so $C_{\{2, 1\}}(G)$ has ≥ 2 isolated vertices. Confirmed by brute force.
21	§6: $B_G \cap [n+1, \infty) = \emptyset$; B_G is a union of $Q \cap I$ over selected components plus a subset of $E_{\{n+1\}}$	VALID	Direct from 19 and Proposition 4.

#	Step	Label	Note
22	§6: $\sup B_G \in E_{\{n+1\}}$	VALID	$B_G \neq \emptyset$ ($2 \in B_G \subseteq E_{\{n+1\}}$ for $n \geq 1$) and $B_G \subseteq [2, n+1]$. If the sup is attained it cannot lie inside an included interval (which contains larger rationals), so it lies in $E_{\{n+1\}}$; if not attained it is the right endpoint of one of the finitely many included intervals, and since $(n+1, \infty)$ is excluded that endpoint is finite, hence in $E_{\{n+1\}}$.
23	§6: $M_C(G) = \sup B_G$	VALID	Writing $\sigma = \sup B_G \geq 2$: every rational $\ell > \sigma$ lies in the defining set (any $k/q \geq \ell > \sigma \geq 2$ is outside B_G , so <i>all</i> its representations mix); every rational $\ell < \sigma$ fails it (pick $r \in B_G$ with $r > \ell$). Hence $\inf = \sigma$ irrespective of the behaviour at $\ell = \sigma$. Also $\sigma \geq \chi_C(G)$, so the source's extra clause $r \geq \chi_C(G)$ is satisfied (see Interpretation).
24	"Scope and exceptional cases" (edgeless graphs, non-integrality, non-attainment)	GAP (harmless)	The $M_C = -\infty$ worry for edgeless graphs is an artefact of OPG's truncated definition; under Brewster–Noel Definition 1.2 ($r \geq \chi_C(G)$) an edgeless graph has $M_C = 1$. The writeup correctly disclaims integrality and attainment.

Summary of labels: 22 VALID, 1 GAP (step 24, cosmetic), 0 ERROR. The only substantive omission is the unchecked equivalence of conventions described under *Interpretation*; I closed it above, and it does not change the theorem.

B.3 Reference check

The writeup invokes **no external theorem at all** — it is fully self-contained (it deliberately re-proves its own $M_C \leq n+1$ instead of citing Brewster–Noel's sharper $\max\{(|V|+1)/2, M(G)\}$). So the reference check reduces to (a) confirming the problem's own definitions and (b) confirming the problem is still open.

1. **Brewster & Noel, "Mixing Homomorphisms, Recolourings, and Extending Circular Precolourings", arXiv:1412.3493** (fetched, text extracted with `pdftotext`). Confirmed:
 - Definition 1.2 as quoted above, including the clause $r \geq \chi_C(G)$ that OPG drops.
 - (k, q) -mixing = $C_{\{k, q\}}(G) = C_{\{G_{\{k, q\}}\}}(G)$ connected, defined for $G \rightarrow H$.
 - **Theorem 4.12:** "For a graph G , $M_C(G) \leq \max\{(|V(G)|+1)/2, M(G)\}$ " — confirmed verbatim, exactly as quoted in `prompt.md`.
 - **§7.1 "Questions for future study", Question 1:** "Is the circular mixing threshold (number) always rational?" — confirmed; also Q2 (integrality) and Q3 (attainment), which the writeup explicitly does *not* claim.
 - Confirmed the $G_{\{19, 7\}}$ remark (" $M_C(G_{\{19, 7\}}) \geq 19/2$ "). This is a useful independent consistency check on the new theorem: $19/2$ has reduced numerator $19 \leq |V(G_{\{19, 7\}})| + 1 = 20$, so the claimed bound is *not* contradicted, and it is close to tight.
 - The paper contains no result asserting that (k, q) -mixing depends only on k/q ; its scaling machinery (Lemmas 4.3/4.6/4.11, "lower parents", (kd, qd) -flexibility, Prop. 4.8) is Farey-based and strictly weaker than the writeup's Lemma 3.
2. **Brewster, McGuinness, Moore, Noel, arXiv:1508.05573** (dichotomy: poly-time for $2 \leq p/q < 4$, PSPACE-complete for $p/q \geq 4$). Fetched; the abstract confirms the dichotomy but says nothing about rationality of M_C or ratio-only dependence.
3. **Brewster & Moore, "Characterizing circular colouring mixing for $p/q < 4$ ", arXiv:2008.12185 / JGT 2023.** Fetched the HTML version. It gives a wind-based characterization for $2 < p/q < 4$ and settles the bipartite $m_C = 2$ conjecture; it does **not** prove rationality of M_C , nor a general local-constancy/finiteness statement.

No citation could be faulted, because none is made; the two definitional facts the writeup silently relies on ($G \rightarrow G_{\{k, q\}} \leftrightarrow k/q \geq \chi_C(G)$), and χ_C rational/attained for finite graphs) are standard and are stated in the source paper's §1 (" $G_{\{k', q'\}} \rightarrow G_{\{k, q\}}$ if and only if $k'/q' \leq k/q$ "; " $\chi_C(G) = \min\{k/q : G \rightarrow G_{\{k, q\}}\}$ ").

Priority. I found no source claiming the result. The problem is listed as open in the source paper's §7.1, on OPG, and in the catalog page's 2026-05 literature review. So this is **not** ALREADY_KNOWN on the evidence available; I cannot of course rule out an unindexed preprint.

B.4 Computational check

Scripts in `verification_astra/scripts/mixing_circular_colourings_0/`. All use exact `Fraction` arithmetic and the convention "mixing = reconfiguration graph non-empty and connected".

(a) **brute.py** — **brute-force mixing and the Theorem's prediction.** Enumerates every (k, q) -colouring, builds the reconfiguration graph, tests connectivity, for all $k \leq K, 1 \leq q \leq k/2$.

graph	n	$E_{\{n+1\}}$	bad ratios (some representation not mixing)	Prop. 4 violations	bad ratio $\geq n+1$
K2	2	{2,3}	{2}	0	none
P3	3	{2,3,4}	{2}	0	none
K3	3	{2,3,4}	all 53 tested rationals in $[2, 23/6]$ ($k \leq 26$), none ≥ 4	0	none
P4, C4, star($K_{\{1,3\}}$)	4	{2,5/2,3,4,5}	{2}	0	none
K4	4	{2,5/2,3,4,5}	all tested rationals in $[2, 9/2]$, none ≥ 5	0	none
C5, bull	5	{2,5/2,3,4,5,6}	all tested in $[2, 11/3]$, none ≥ 4	0	none
K23	5	{2,5/2,3,4,5,6}	{2}	0	none

“Prop. 4 violations” counts ordered pairs of tested ratios lying in the *same* component interval of $(2, \infty) \setminus E_{\{n+1\}}$ with different mixing status: **0 in every case**, over $k \leq 13$ (and $k \leq 26$ for $K3/P3$). No ratio was ever representation-dependent in range. No counterexample to §5 ($k/q \geq n+1 \Rightarrow$ mixing).

Resulting thresholds, all in $E_{\{n+1\}}$ as the Theorem requires: $Mc(K2)=2$ (matches the value quoted on OPG), $Mc(P3)=Mc(P4)=Mc(C4)=Mc(K_{\{1,3\}})=Mc(K_{\{2,3\}})=2$, $Mc(K3)=4$, $Mc(K4)=5$, $Mc(C5)=Mc(bull)=4$.

(b) **lemmas.py — direct tests of Lemmas 1–3 and the quotient criterion.** For $K2, P3, K3, P4, C4, paw, K4$ and all $k \leq 12$: * Lemma 2 failures (a non-empty class X_D disconnected) at $k/q \notin E_n$: **none**, in any case, under either unit moves or arbitrary single-vertex recolourings. * Lemma 2 failures at $k/q \in E_n$: present exactly as the writeup predicts and needs — e.g. $K2$ at $(4,2), (6,3), (8,4), (10,5)$ (ratio 2), $K3$ at $(6,2), (9,3), (12,4)$ (ratio 3), $K4$ at $(8,2), (12,3)$ (ratio 4), $C4$ at ratios 2 and 4. So E_n is genuinely necessary and the writeup is right to quarantine it. * Quotient-criterion failures (R connected $\neq Q$ connected) at $k/q \notin E_n$: **none**. * Lemma 3(1) (the set of feasible orientation classes depends only on k/q , not on the representation): **no violation**, i.e. the integrality trick of step 11 checks out empirically. * Lemma 3(2) (that set is constant on component intervals of $(2, \infty) \setminus E_{\{n+1\}}$): **no violation**.

(c) **endtoend.py — decisive test.** I re-implemented the writeup’s *algorithm* from scratch: build $Q_{\{k,q\}}(G)$ using nothing but Bellman–Ford negative-cycle detection on the *-augmented difference-constraint digraphs of §3 (class feasibility on n vertices, transition feasibility on the $n+1$ -vertex twin graph of §4), then answer “non-empty and connected”. Compared against brute-force mixing for $K2, P3, K3, P4, C4, paw, K4, diamond, C5, bull, house, K23$ and all $k \leq 11$:

360 (k,q) pairs tested; 0 mismatches — including 0 mismatches even at the exceptional ratios in E_n , where the theory only promises agreement off E_n .

This simultaneously validates Lemma 1, Lemma 2, Lemma 3 and the §4 quotient reduction as an executable specification.

(d) **threshold.py — larger graphs via the validated algorithm** ($k \leq 16$):

graph	n	max bad ratio	min good ratio above it	$E_{\{n+1\}}$ point forced to be the threshold	Prop. 4 violations	rep.-dependent ratios
C6 (= L_3)	6	15/4	4	4	0	none
C7	7	15/4	4	4	0	none
prism($K_3 \square K_2$)	6	15/4	4	4	0	none
K5	5	11/2	6	6	0	none

Every observed threshold is in $E_{\{n+1\}}$, no ratio was representation-dependent, and no bad ratio $\geq n+1$ appeared. $Mc(C6) = 4$ agrees with Brewster–Noel’s $M(L_m) = m+1$ combined with their Theorem 4.12 ($Mc(L_3) \leq \max\{7/2, 4\} = 4$).

Tightness observation. $Mc(K_n) = n+1$ for $n = 2, 3, 4, 5$ (values 3? no — 2, 4, 5, 6; note $Mc(K_2) = 2 < 3$ is the known K_2 exception, while $Mc(K_3)=4, Mc(K_4)=5, Mc(K_5)=6$ all equal $n+1$). Hence the §5 bound $Mc \leq n+1$ is attained and the index $n+1$ in $E_{\{n+1\}}$ cannot be lowered to E_n : the theorem’s constant is sharp in that sense.

No computational check failed.

B.5 Caveats

- Convention/definition gap (the one real omission).** The writeup adopts (i) “mixing = non-empty *and* connected” and (ii) an infimum over *all* rationals ℓ , both inherited from OPG’s rendering, which drops Brewster–Noel’s clause $r \geq \chi_C(G)$. It never verifies that this yields the same number as the source definition. It does (proof under *Interpretation*), but the check is load-bearing for the claim’s relevance and is missing from the writeup.
- Edgeless graphs.** The final paragraph’s $Mc = -\infty$ scenario does not arise under the real definition ($\chi_C = 1, Mc = 1$). The theorem’s hypothesis “at least one edge” is therefore unnecessary, and the writeup’s framing of this as “a trivial normalization issue in the quoted statement” over-dramatizes a non-issue.
- mc not covered.** Brewster–Noel’s Question 1 is phrased for “the circular mixing threshold (number)”, i.e. also for $mc(G)$. The writeup silently restricts to Mc . The same machinery yields $mc(G) \in E_{\{n+1\}}$ but this is neither stated nor proved.
- Nothing about integrality or attainment** — correctly disclaimed in the verdict block’s caveats. In particular the writeup does **not** answer OPG’s companion questions (“Is Mc always an integer?”, “Is Mc attained for non-bipartite graphs?”). Its result is consistent with, and does not settle, the observation that no non-integer example is known. Note that a non-integer value is *permitted* by the theorem (e.g. $19/2$ for $G_{\{19,7\}}$ would be allowed).

5. $E_n \subseteq E_{n+1}$ is used in Proposition 4 without comment; trivially true.
6. **Simple loopless finite graphs only.** The writeup says “finite loopless”; multigraphs are harmless (parallel edges give duplicate constraints), but loops make the colouring set empty and are correctly excluded. Directed/infinite graphs are out of scope, as they are for the original problem.
7. **The numerator bound is on the reduced numerator only.** The statement gives no bound on the denominator, hence no effective bound on the size of $M_C(G)$ beyond $\leq n+1$ (which §5 gives anyway). The writeup does not claim more.
8. **Sharpness of $n+1$ vs Brewster–Noel’s $\max\{(|V|+1)/2, M(G)\}$.** §5’s self-contained bound is much weaker than the published one; this is a deliberate, harmless choice (it keeps the proof independent), but it means the paper does not recover or improve the known bound. Combining the numerator bound with Theorem 4.12 would in fact give a stronger conclusion than either alone.
9. **Verification limits.** All computations are on graphs with $n \leq 7$ and $k \leq 26$. The lemmas were verified as executable specifications, not proved by machine; the proofs themselves I checked by hand and found sound.

B.6 Referee summary

I set out to break this and could not. The architecture is: (1) partition the (k, q) -colourings by the edge-orientation they induce, at which point each class is exactly the integer-point set of a system of difference constraints with arc weights only $-q$ and $k-q$; (2) observe that such a system’s unit-step graph is connected unless there is a zero-weight simple cycle, which pins k/q to a rational b/a with $b \leq n$; (3) observe that *feasibility* of such a system — including the inhomogeneous box $0 \leq x \leq k-1$ — is a finite boolean combination of comparisons of k/q with rationals of numerator $\leq n+1$, the inhomogeneity being killed by the integrality identity $ak - bq - 1 \geq 0 \Leftrightarrow ak - bq > 0$; (4) reduce mixing to connectivity of a quotient graph on orientations whose vertices and edges are exactly such feasibility tests, the edge tests living on an $(n+1)$ -vertex twin graph. Every one of these steps re-derived correctly by hand, and I re-implemented the whole pipeline independently and matched brute-force mixing on 360 parameter pairs across 12 graphs with zero mismatches, plus threshold computations on four 5–7-vertex graphs that all landed in E_{n+1} . The exceptional set E_n is genuinely needed (orientation classes do disconnect at $k/q \in E_n$ — e.g. K_4 at $(8, 2)$ and $(12, 3)$), and the writeup correctly quarantines those ratios rather than assuming them away; the bound $n+1$ is sharp, since $M_C(K_n) = n+1$ for $3 \leq n \leq 5$. No supremum is assumed attained anywhere — indeed the argument is carefully built so that the behaviour *at* the exceptional ratios is irrelevant to the supremum, which is exactly the trap the claim type invites. The single genuine defect is that the writeup formalizes M_C from OpenProblemGarden’s rendering, which truncates Brewster–Noel’s Definition 1.2 by dropping the clause $r \geq \chi_C(G)$, and then patches it with an unverified non-emptiness convention; the two readings do coincide, and I proved they do, but the writeup owes that half-page. Add that, delete the misleading edgeless-graph paragraph, and (optionally) extend the one-line argument to the circular mixing *number* m_C , and this is a correct and publishable resolution of Brewster–Noel Question 7.1(1) / OPG “Mixing Circular Colourings”, strictly stronger than the question asked. **MINOR_GAPS**, high confidence.

References

- [1] R. C. Brewster and J. A. Noel, *Mixing Circular Colourings*, OpenProblemGarden, posted 22 September 2012, http://www.openproblemgarden.org/op/mixing_circular_colourings_0.
- [2] R. C. Brewster and J. A. Noel, Mixing homomorphisms, recolourings, and extending circular precolourings, *Journal of Graph Theory* **80** (2015), 173–198; arXiv:1412.3493.
- [3] R. C. Brewster, S. McGuinness, B. Moore and J. A. Noel, A dichotomy theorem for circular colouring reconfiguration, *Theoretical Computer Science* **639** (2016), 1–13; arXiv:1508.05573.
- [4] R. C. Brewster and B. Moore, Characterizing circular colouring mixing for $p/q < 4$, *Journal of Graph Theory* **102** (2023), 296–324; arXiv:2008.12185.